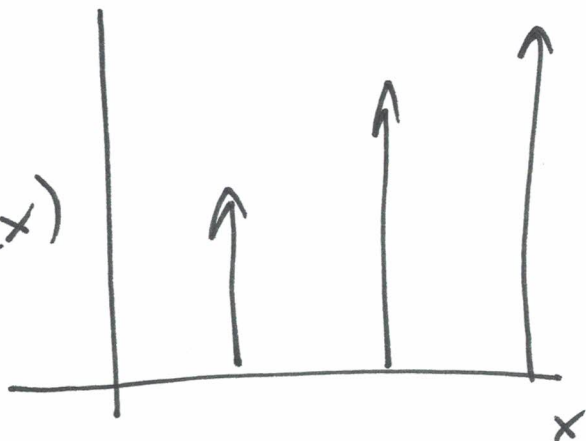


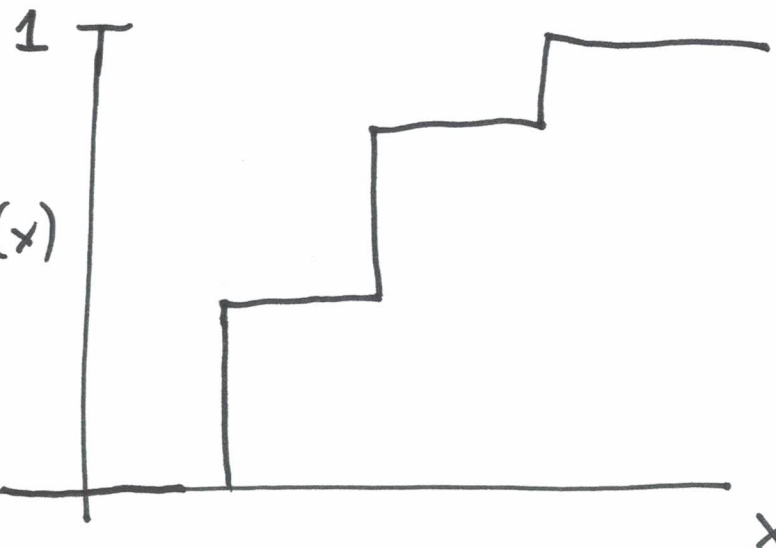
PDF : PROBABILITY DENSITY FUNCTION

DISCRETE

$P_X(x)$

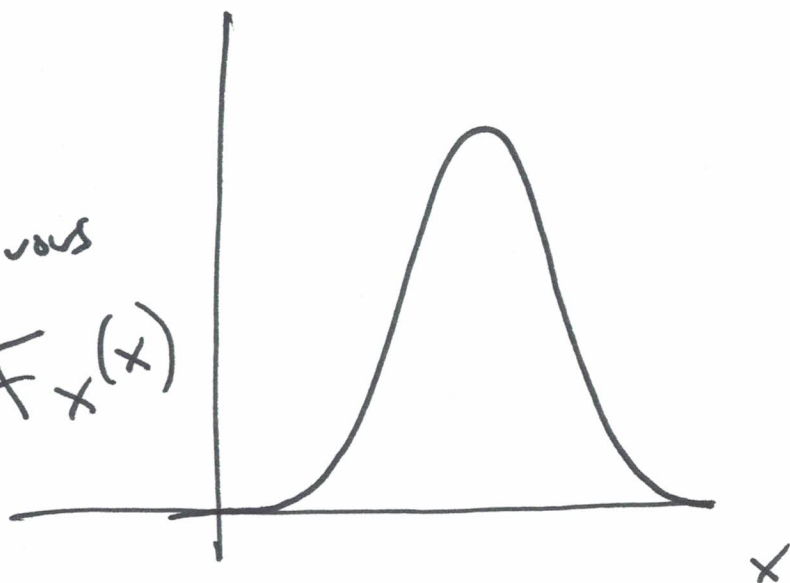


$F_X(x)$

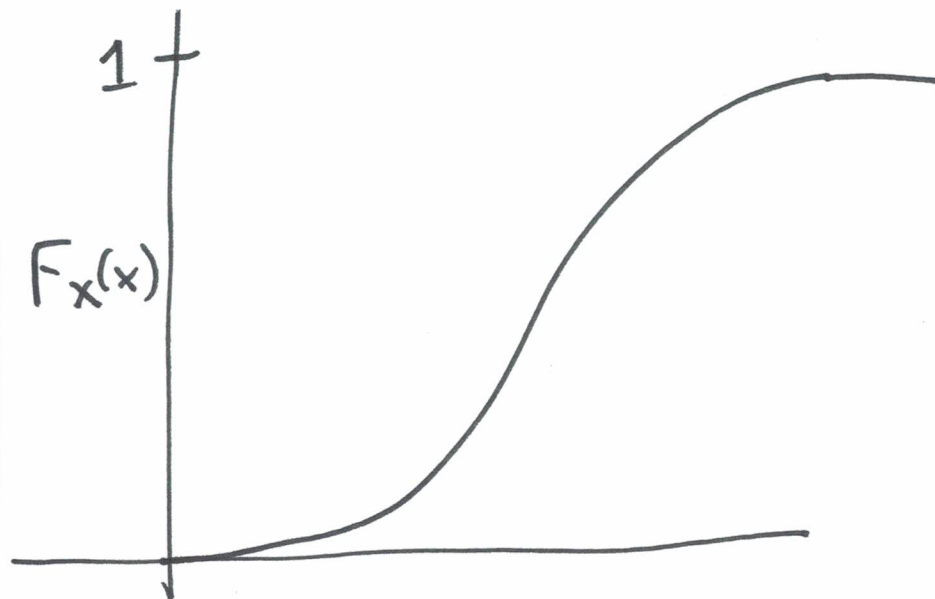


CONTINUOUS

$F_X(x)$



$F_X(x)$



LAST TIME

$$\text{CDF: } F_X(x) = P(X \leq x)$$

THE PDF IS THE DERIVATIVE OF
THE CDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Diagram annotations:

- A red arrow points from the word "SMALL" to the x in the $f_X(x)$ term.
- A red arrow points from the word "BIG" to the x in the $F_X(x)$ term.
- A red arrow points from the word "LITTLE x (SOME VALUE)" to the x in the $F_X(x)$ term.
- A red arrow points from the word "BIG x (NAME OF RV)" to the x in the $F_X(x)$ term.

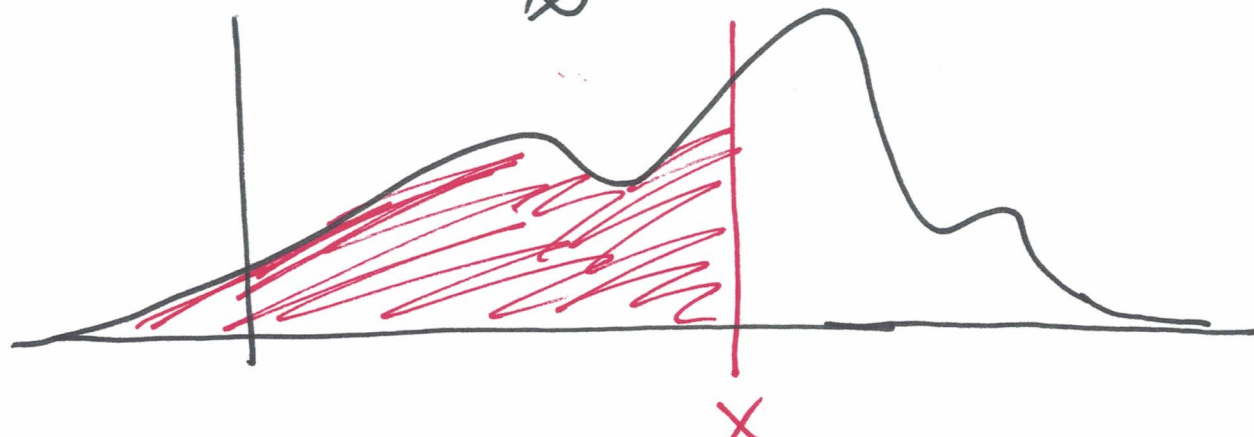
THE CDF IS THE INTEGRAL OF
THE PDF:

$$F_X(y) = \int_{-\infty}^y f_X(x) dx$$

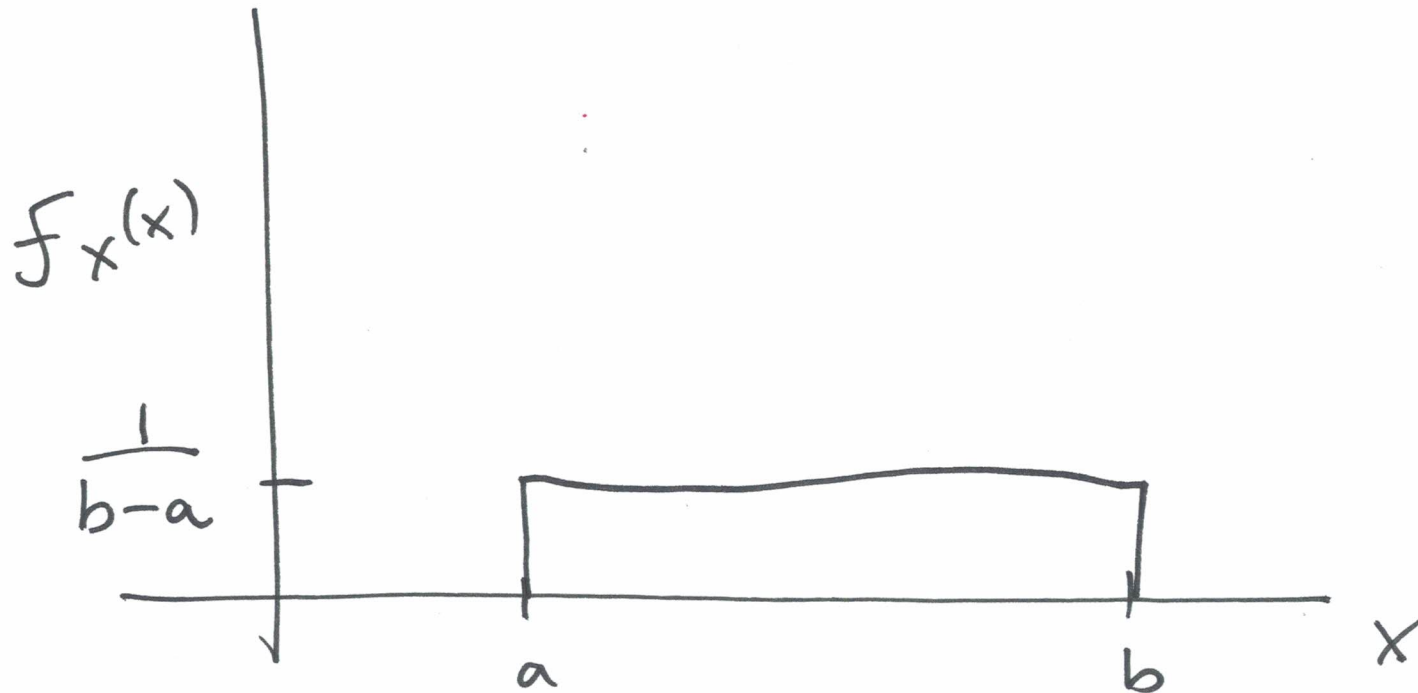
MATCH

PLACEHOLDER FOR INTEGRATION

$$F_X(x) = \int_{-\infty}^x f_X(y) dy$$



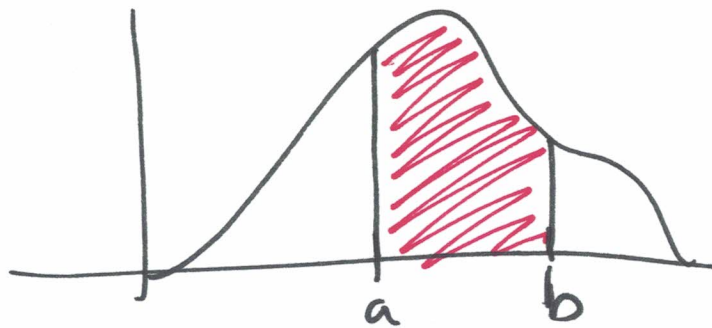
PDF For UNIFORM DISTRIBUTION (CONTINUOUS)
ON $[a, b]$



$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{ELSE} \end{cases}$$

WE USE THE PDF TO DETERMINE
PROBABILITIES OF EVENTS BY INTEGRATING:

$$P(X \in [a, b]) = \int_a^b f_X(x) dx$$

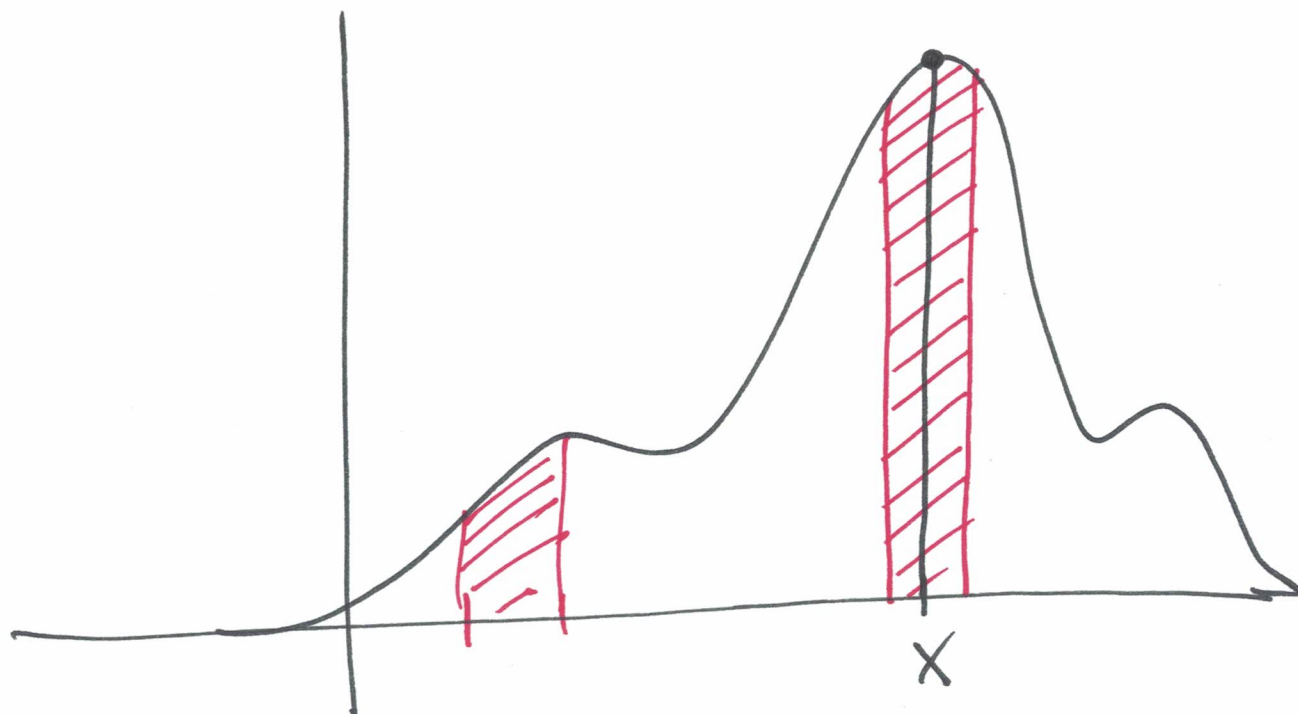


$$F_X(b) - F_X(a)$$



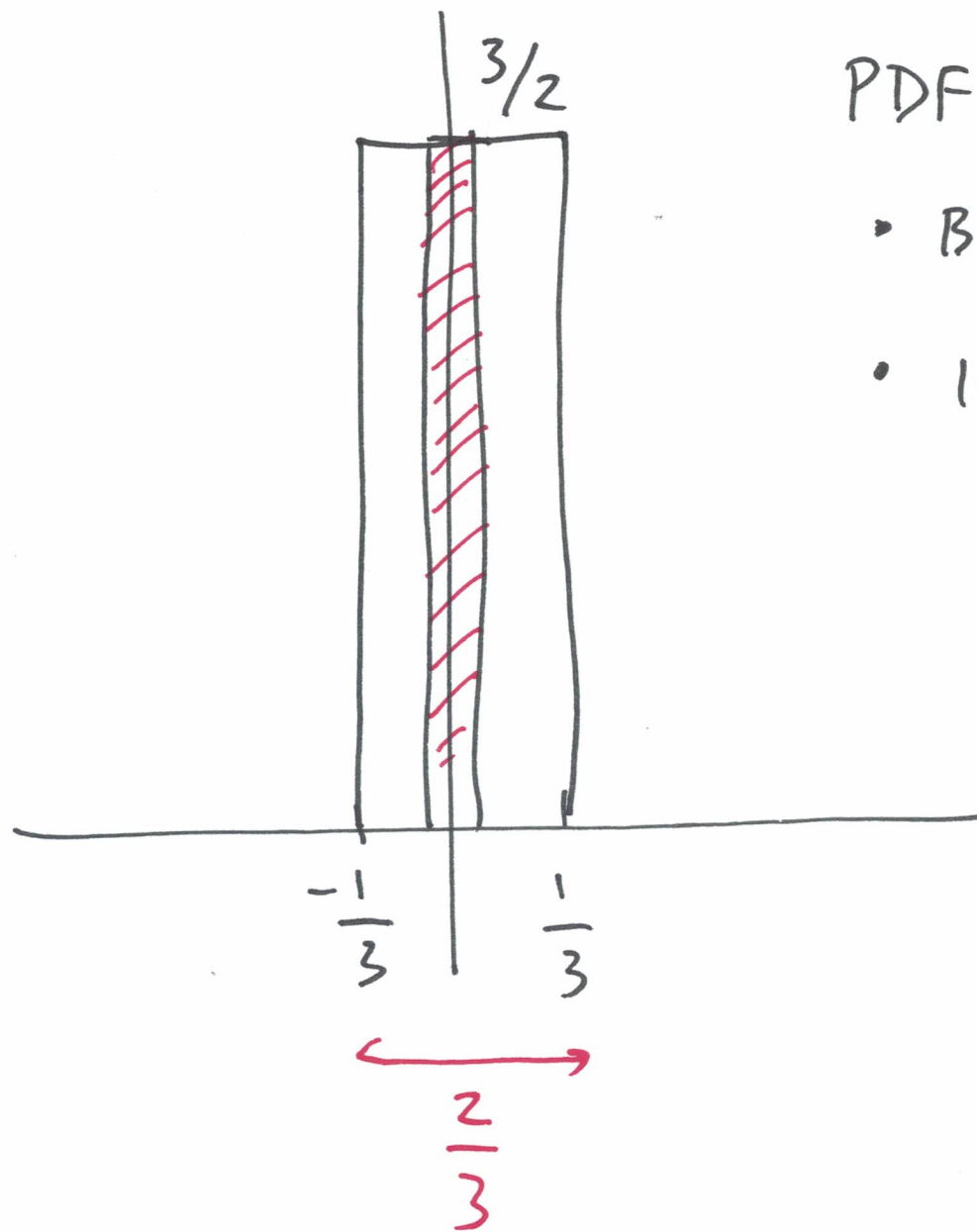
PDF AT x IS NOT

"THE PROBABILITY OF GETTING OUTCOME x "



WE CAN ONLY TALK ABOUT THE
PROBABILITY OF INTERVALS.

EX UNIFORM ON $[-\frac{1}{3}, \frac{1}{3}]$

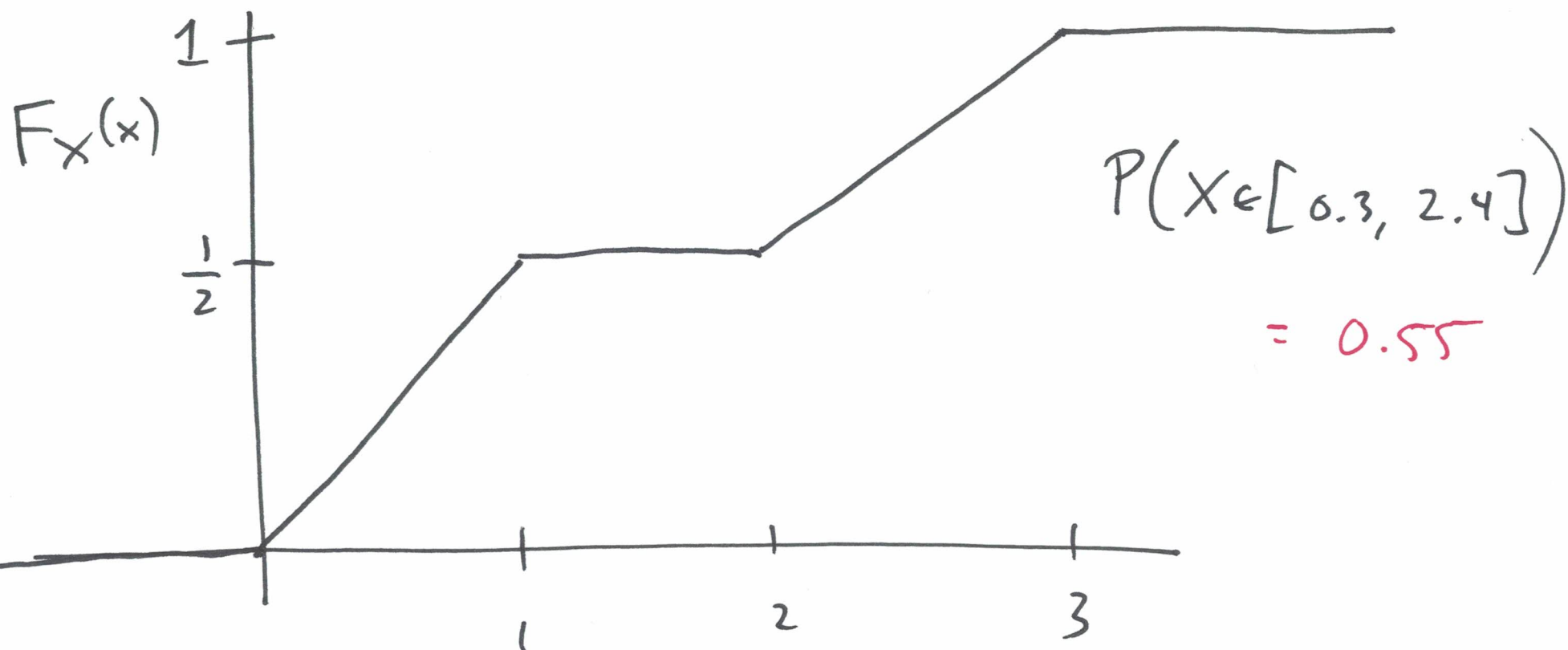
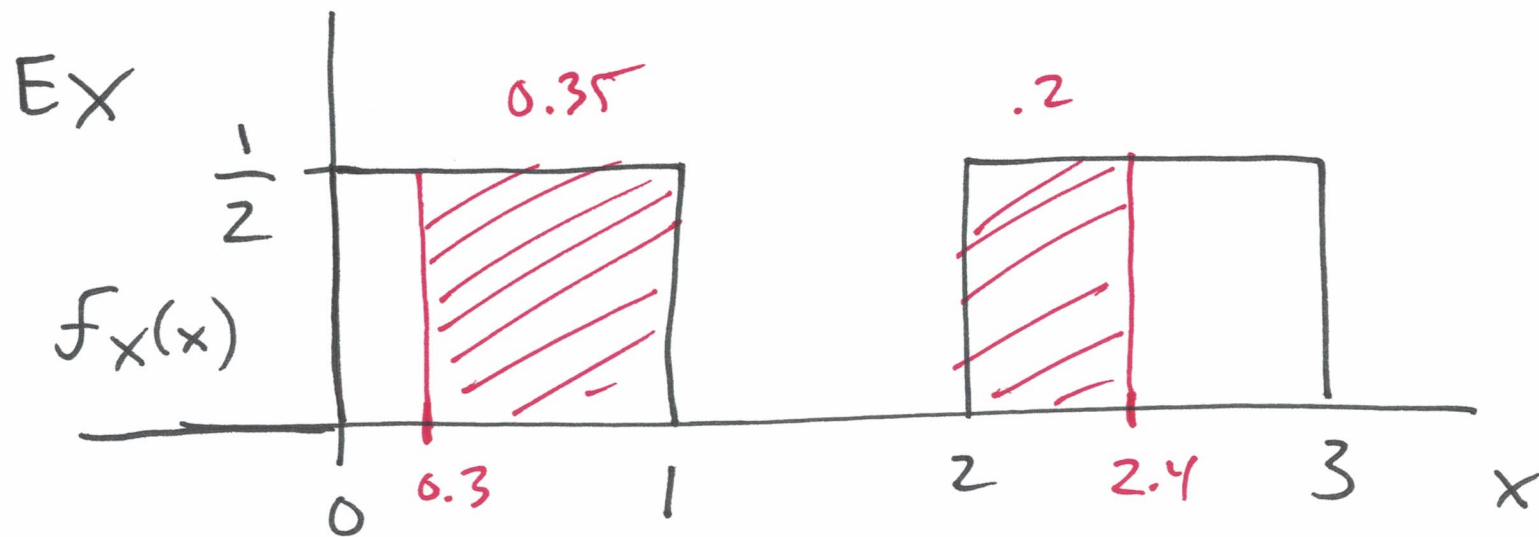


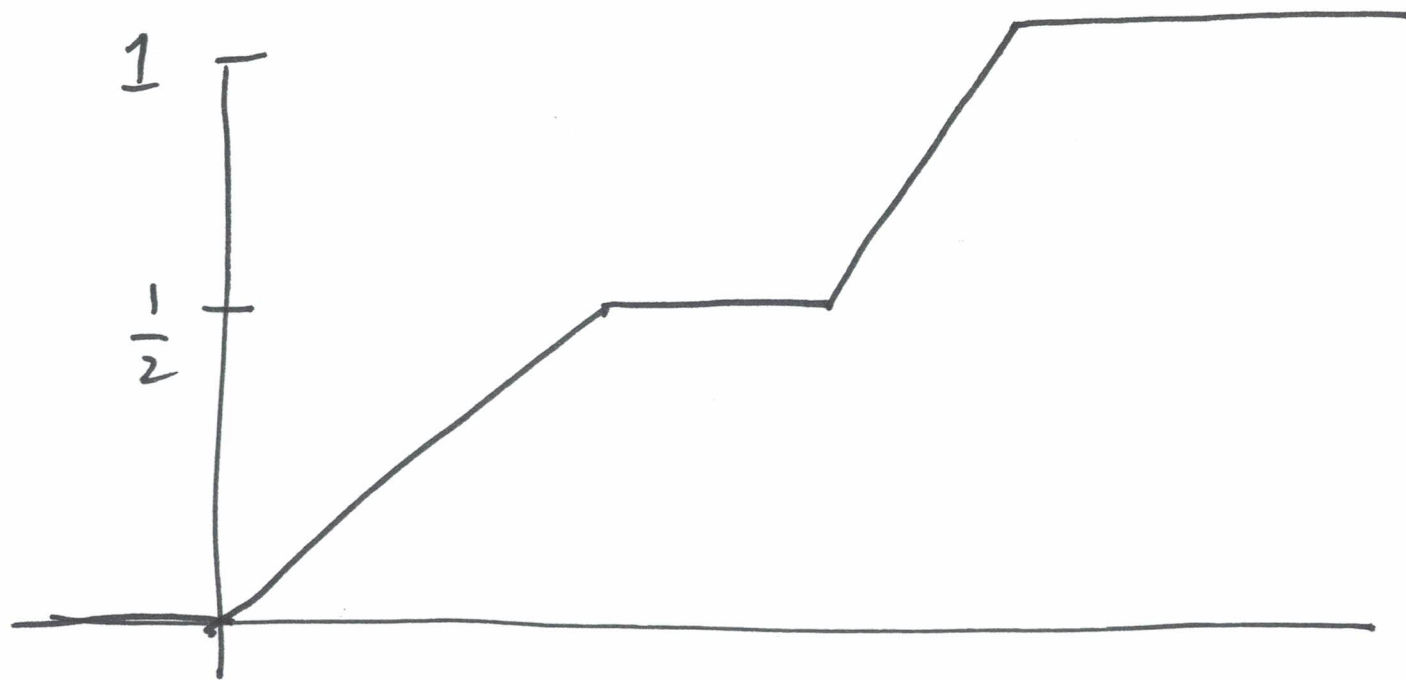
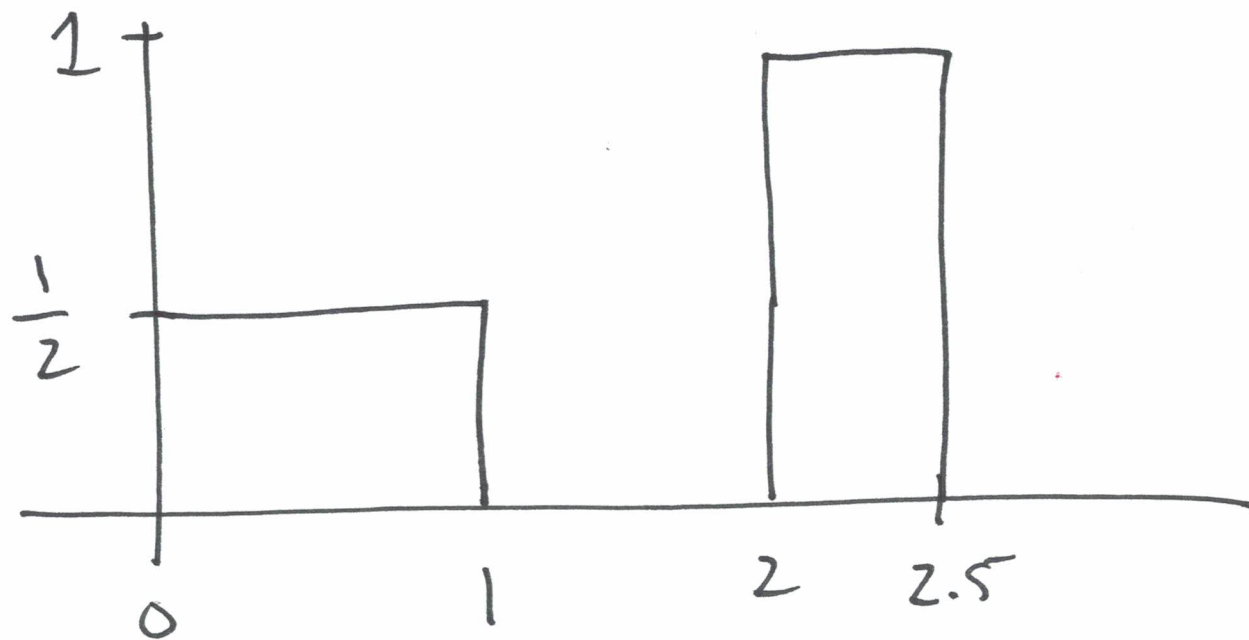
PDF MUST

- BE ≥ 0

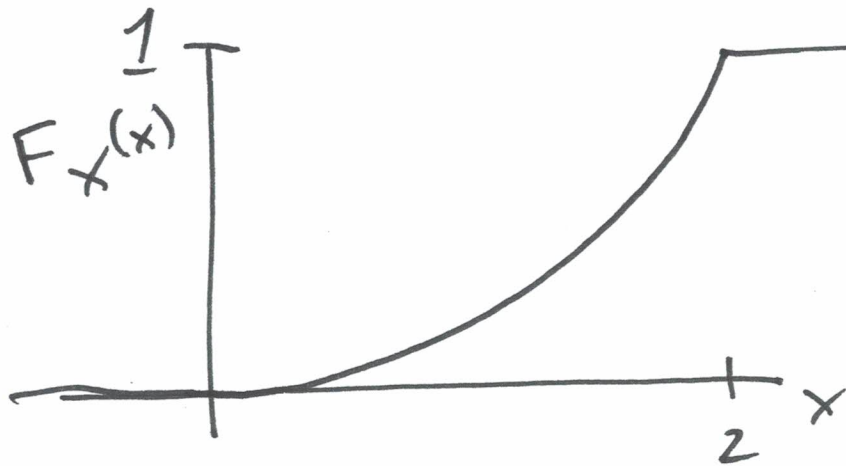
- INTEGRATE TO 1.

$$\int_{-\infty}^{\infty} f_X(x) = 1.$$

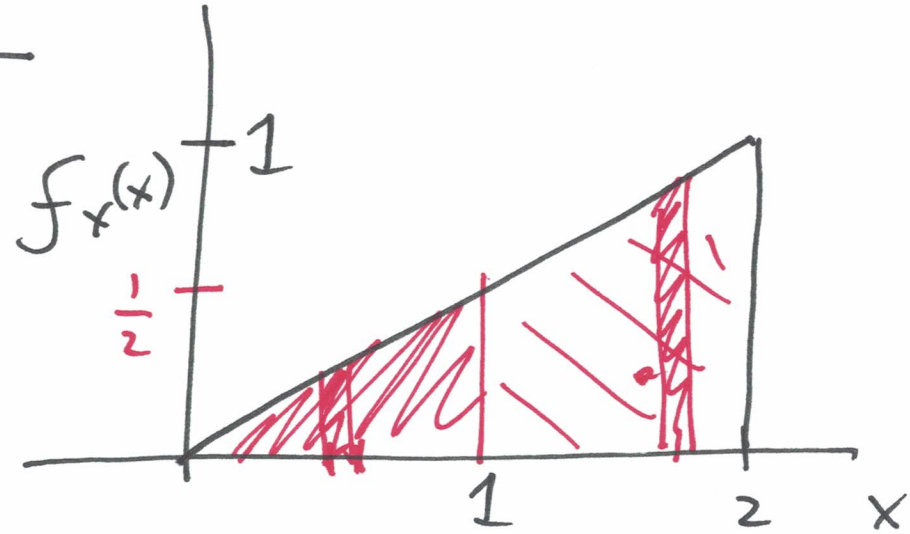




$$F_X(x) = \begin{cases} \frac{1}{4}x^2 & x \in [0, 2] \\ 0 & x < 0 \\ 1 & x > 2 \end{cases}$$



$$f_X(x) = \begin{cases} \frac{1}{2}x & x \in [0, 2] \\ 0 & \text{OTHERWISE} \end{cases}$$



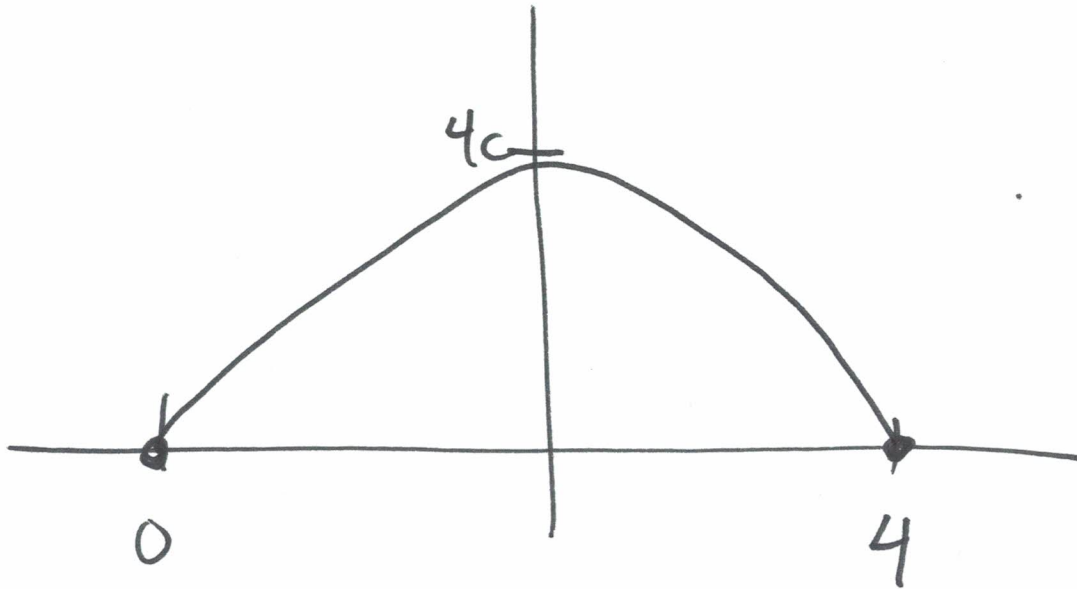
↓

$$P(X < 1) = \frac{1}{4}$$

$$P(X > 1) = \frac{3}{4}$$

$$f_X(x) = \begin{cases} c(4x - x^2) & x \in [0, 4] \\ 0 & \text{ELSE} \end{cases}$$

WHAT VALUE OF c MAKES THIS A PDF?

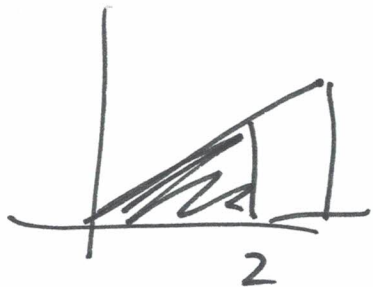


$$\int_0^4 f_X(x) dx = 1.$$

$$\int_0^4 c(4x - x^2) dx$$

$$= c \left[2x^2 - \frac{1}{3}x^3 \right]_{x=0}^{x=4}$$

$$= c \left(\frac{96}{3} - \frac{64}{3} \right) \quad c = \frac{3}{32}.$$

$$\textcircled{1} f_X(x) = \begin{cases} mx + b & x \in [0, 3] \\ 0 & \text{ELSE} \end{cases}$$


WHAT VALUES OF
m, b MAKE THIS A PDF?

$$\textcircled{2} P(X < 2) = \frac{1}{2}$$

$$\int_0^3 mx + b \, dx = 1 \Rightarrow \left[\frac{1}{2}mx^2 + bx \right]_0^3 = \frac{9}{2}m + 3b = 1.$$

$$\int_0^2 mx + b \, dx = \frac{1}{2} = \left[\frac{1}{2}mx^2 + bx \right]_0^2 = 2m + 2b = \frac{1}{2}$$

$$- 3m + 3b = \frac{3}{4}$$

$$\frac{3}{2}m = \frac{1}{4}$$

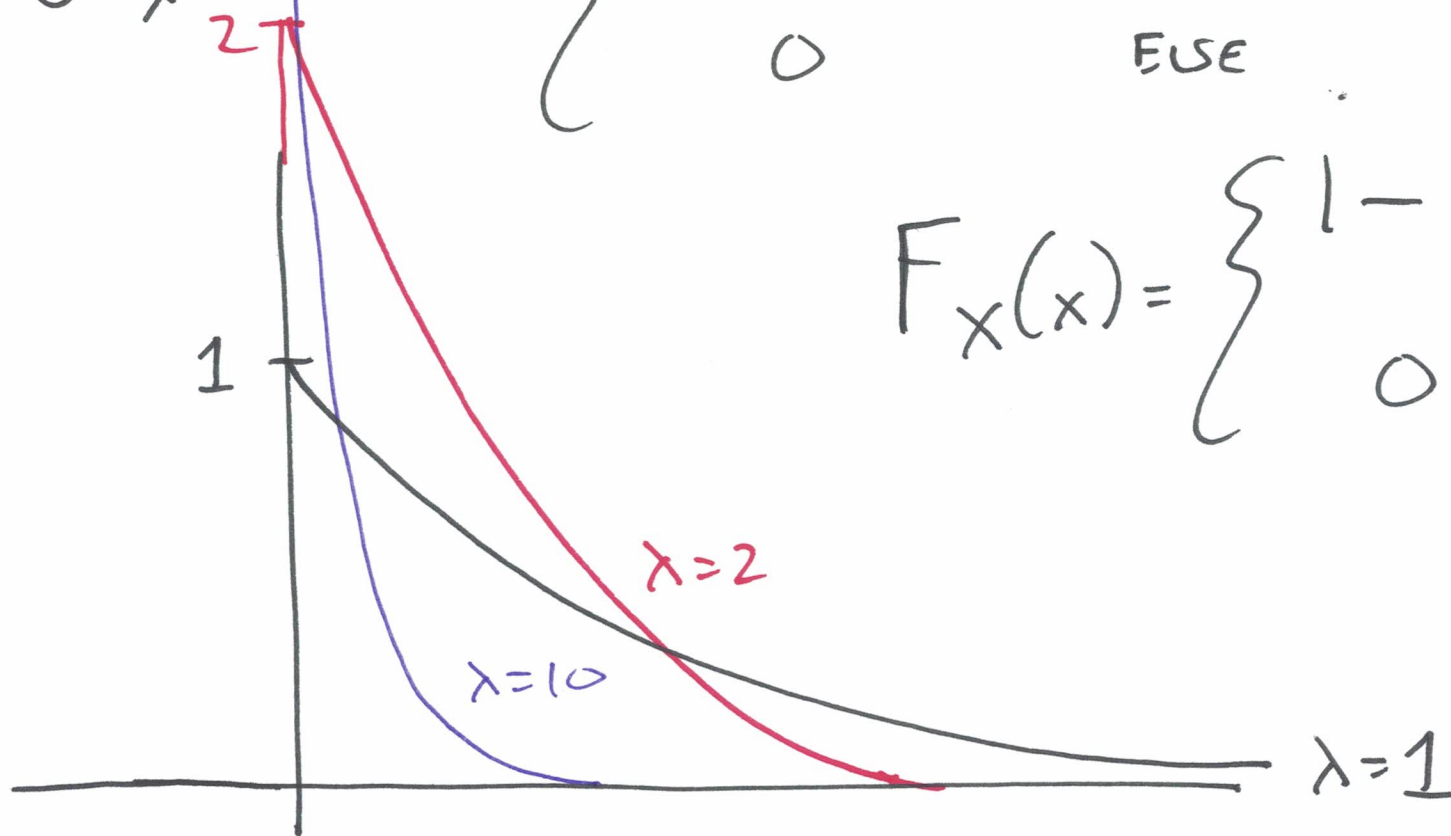
$$\boxed{m = \frac{1}{6} \quad b = \frac{1}{12}}$$

CERTAIN COMMON PDFS HAVE NAMES:

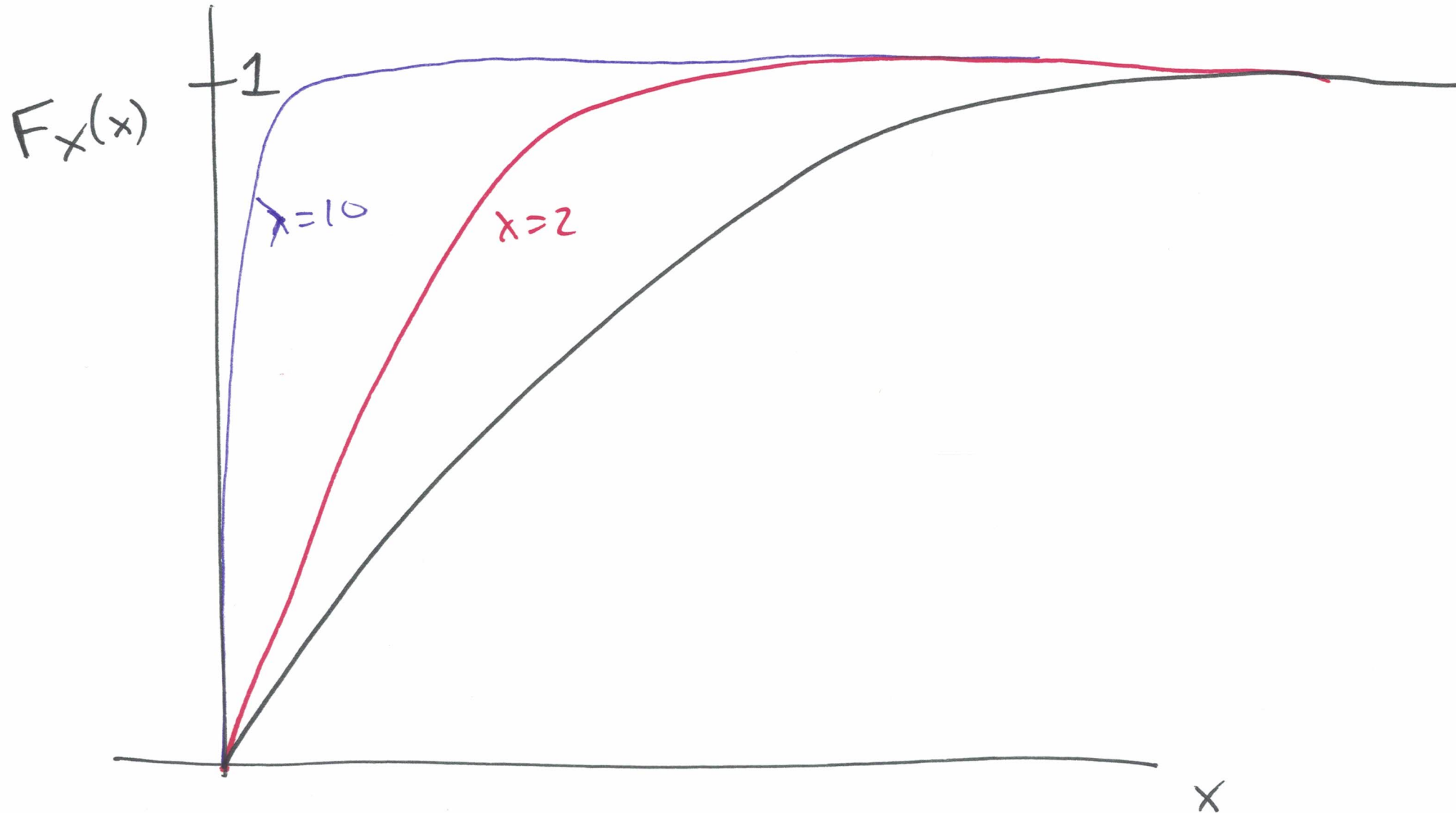
EXPONENTIAL RANDOM VARIABLE:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{ELSE} \end{cases}$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$



EXPONENTIAL CDF



THINK ABOUT ARRIVALS IN TWO WAYS:



How MANY ARRIVALS N IN AN INTERVAL of
LENGTH t ? POISSON RANDOM VARIABLE

How LONG TO WAIT BEFORE NEXT
ARRIVAL? EXPONENTIAL RANDOM VARIABLE.

CONNECTED.

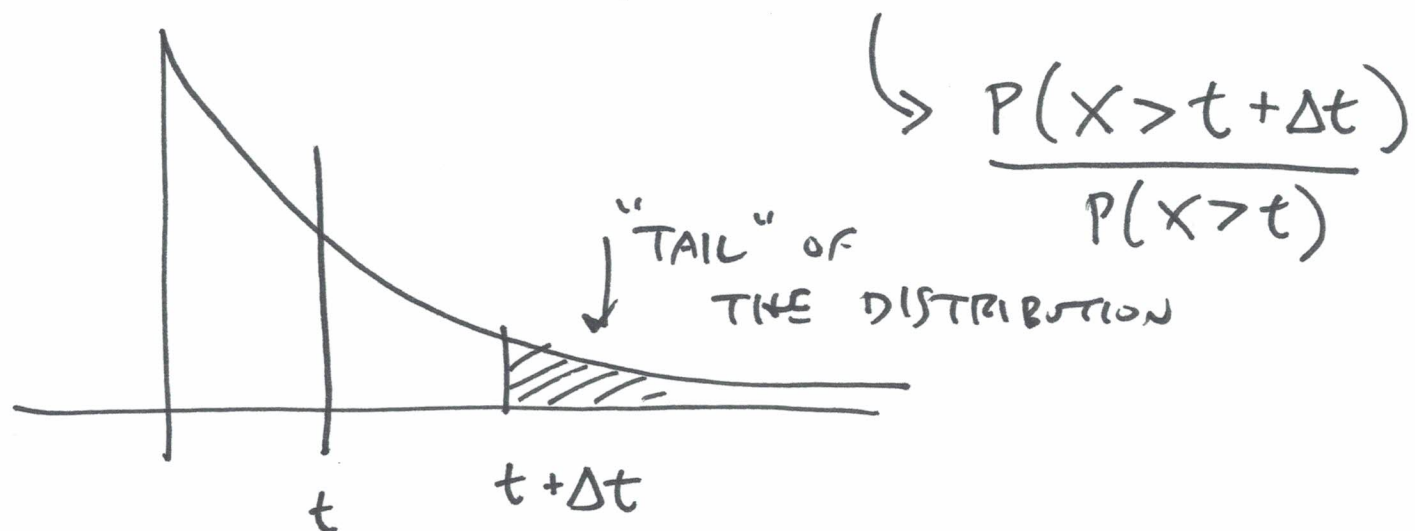
IF THE TIME BETWEEN EVENTS IS
MODELED AS EXPONENTIAL WITH PARAMETER
 λ , THEN THE NUMBER OF ARRIVALS
IN AN INTERVAL OF LENGTH t IS
A POISSON RANDOM P.V. WITH $\alpha = \lambda t$.

$$P(N = k) = e^{-\boxed{\lambda t}} \frac{(\boxed{\lambda t})^k}{k!} \quad k = 0, 1, 2, \dots$$


OTHER FACT ABOUT EXPONENTIAL:

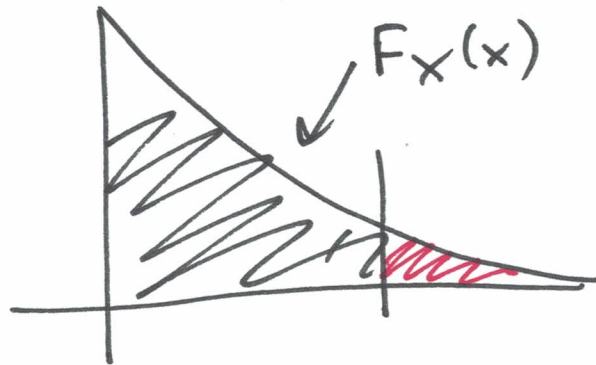
THIS IS THE ONLY CONTINUOUS
RANDOM VARIABLE TO HAVE THE
MEMORYLESS PROPERTY

$$P(X > t + \Delta t \mid X > t) = P(X > \Delta t)$$



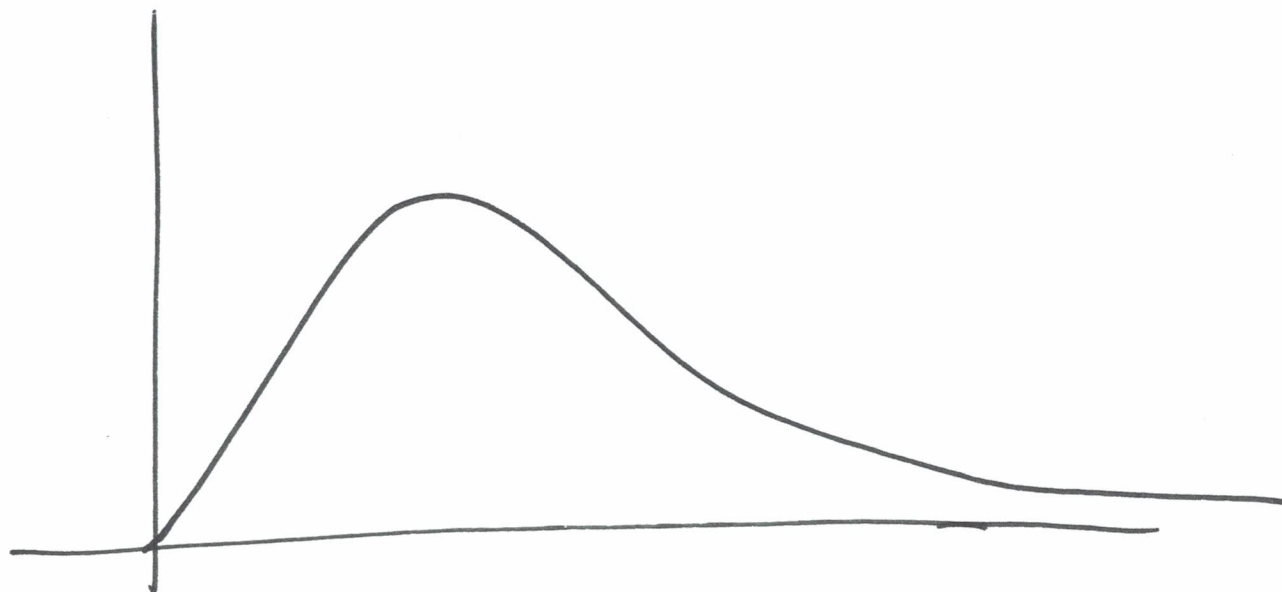
EXPONENTIAL CDF:

$$F_X(x) = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & \text{ELSE} \end{cases}$$

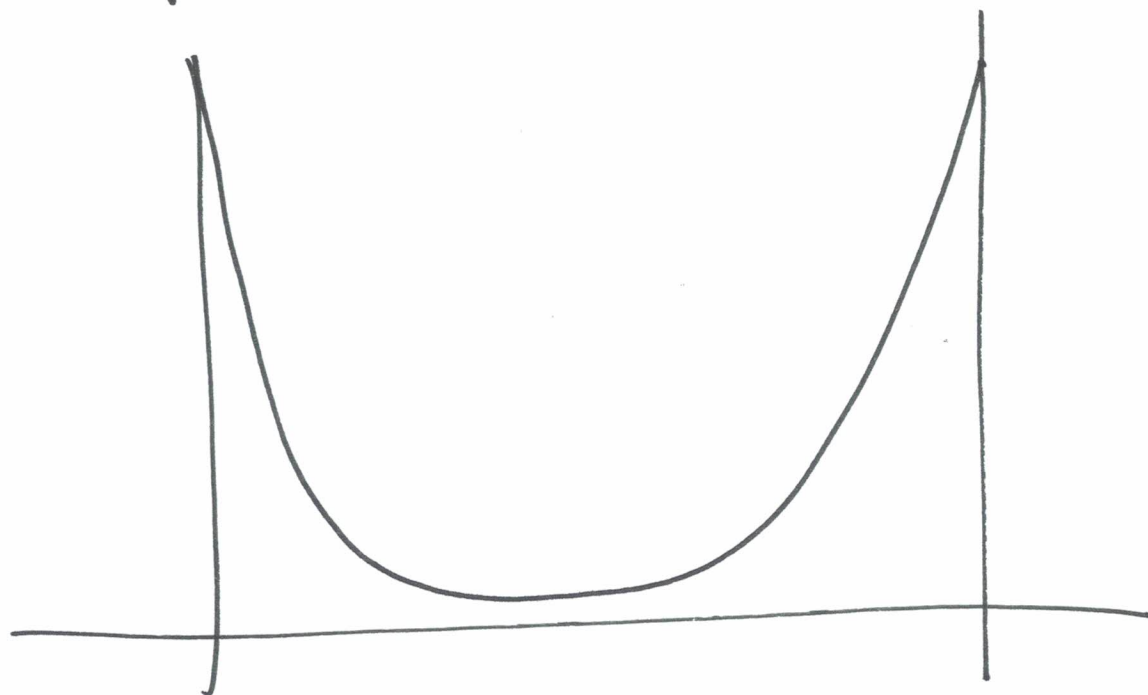


$$P(X > x) = \begin{cases} e^{-\lambda x} & x \geq 0 \\ 0 & \text{ELSE.} \end{cases}$$

$$\begin{aligned} &P(X > t + \Delta t \mid P(X > t)) \\ &= \frac{e^{-\lambda(t + \Delta t)}}{e^{-\lambda t}} = e^{-\lambda \Delta t} = P(X > \Delta t) \end{aligned}$$

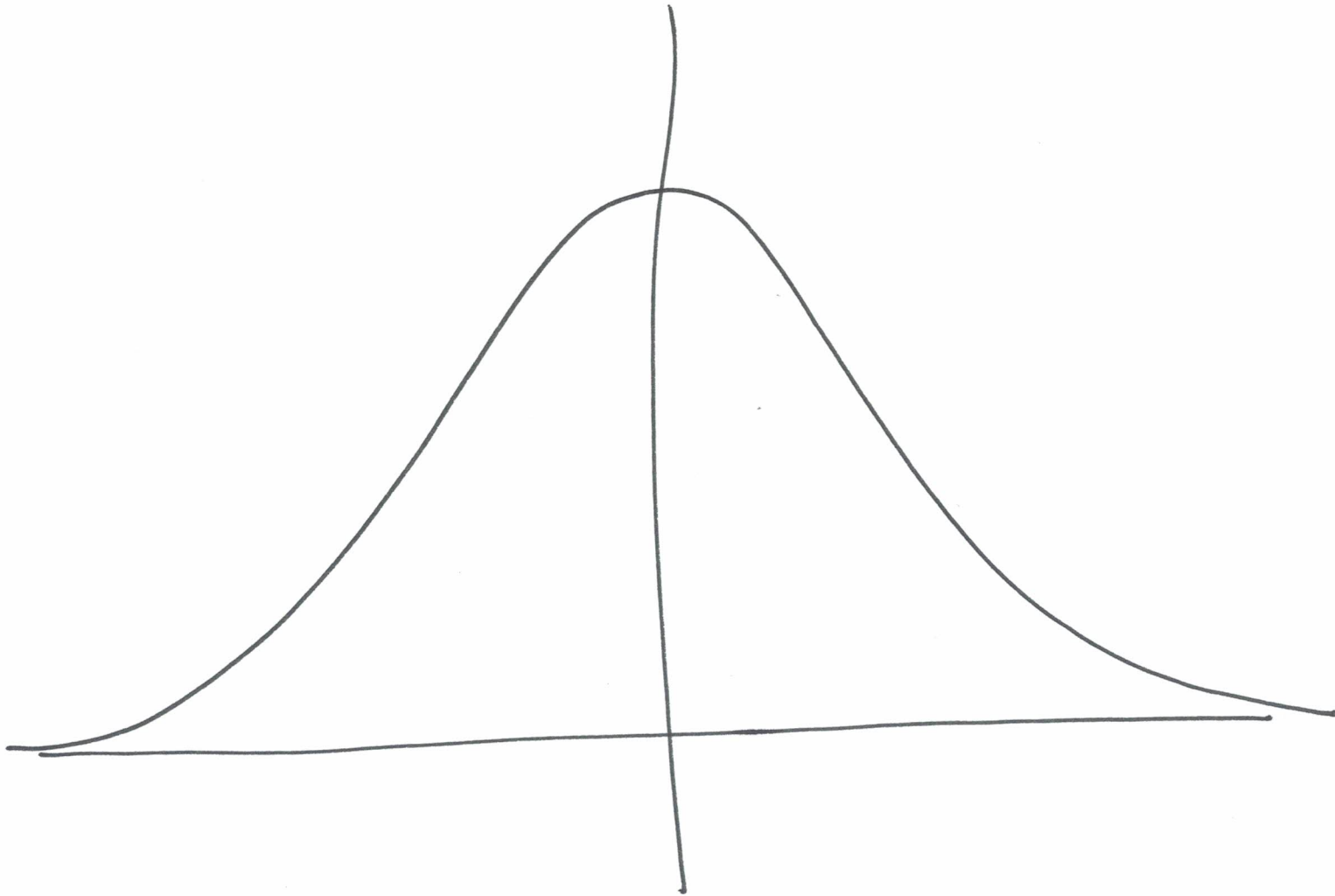


GAMMA



BETA

(*) GAUSSIAN



THE MOST IMPORTANT CONTINUOUS R.V.

EX OF MEMORYLESS.

SAY THAT RPI SHUTTLE INTERARRIVAL
TIMES ARE EXPONENTIAL w/ $\lambda = \frac{1}{20}$.

SAY WE WAITED 15 MIN FOR A
SHUTTLE ALREADY. WHAT'S THE PROB
WE NEED TO WAIT AT LEAST
25 MIN TOTAL?

MEMORYLESS $\Rightarrow$ LIKE WAITING 10 MIN TO

BEGIN WITH.

$$P(X > 10) = e^{-\frac{1}{20} \cdot 10} \\ = e^{-\frac{1}{2}}.$$